

Flavor profiles as sparse condition vectors

MolPallette — what was measured, and why the approach was not adopted

SUMMARY

MolPallette's brief specifies a condition vector that is either protein-pocket context or neutral. This report records what happened when we tried to build a flavor-derived condition signal for the 393,066-molecule corpus, and why every route examined was rejected.

The conclusion is negative and the evidence is quantitative. Flavor labels do not exist at corpus scale, cannot be derived from chemical class or source organism at usable precision, and cannot be imputed by a frontier LLM at a rate that beats a free deterministic baseline. The pretrained model is therefore UNCONDITIONAL, and that is the recommended state.

One prior failure frames all of it. An earlier 97-bit RDKit fragment condition vector was found to LEAK: it was computed from the intact molecule G, which contains the very R-group the retrieval objective must predict. Removing it took rank1_distinct from 680 to 5,851 and coverage from 11.4% to 100%. Every candidate below is judged first on whether it repeats that mistake.

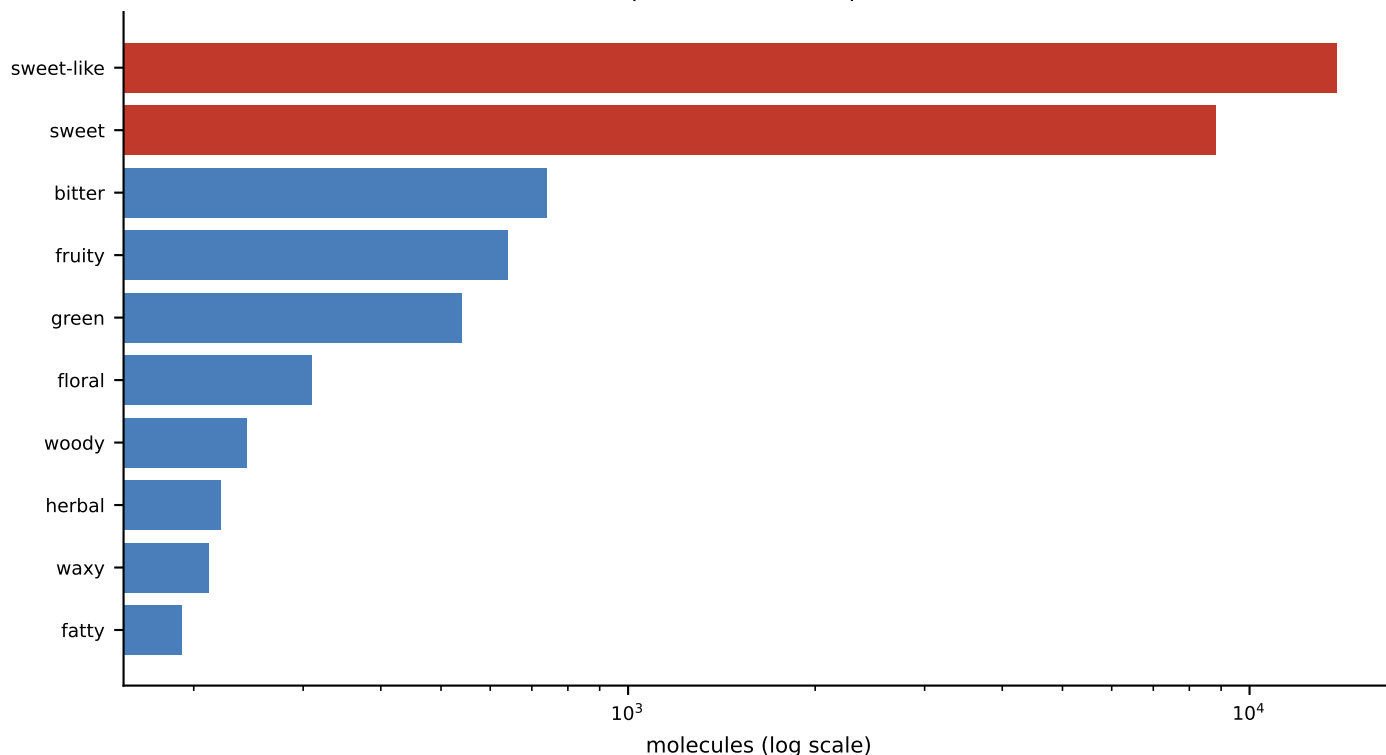
ROUTE	VERDICT
measured labels (FlavorDB)	5.29% corpus coverage; usable only for fine-tuning, not conditioning
InChIKey join to COCONUT	1.47% ceiling, dominated by an imputed class
source organism	tested, fails: mango is 4% "fruity"
chemical class	tested, fails: alkaloid->bitter is 19% precise
LLM annotation	benchmarked: loses to the class prior at n=300
physical exclusion (MW/TPSA)	VALID, and free: ~209K honest odorless labels

Prepared from measurements run against coconut-flavordb_v5/v6 and FlavorDB 2026-08.

1. The label vocabulary is degenerate

FlavorDB 2026-08: 25,106 labelled molecules, 33,785 descriptor assignments

top 10 of 715 descriptors



715 distinct descriptors, but the distribution is pathological:

sweet-like + sweet	22,593 assignments	= 66.9% of all mass
appear $\geq 100\times$	24 descriptors	
appear $\geq 10\times$	196	
appear exactly once	253	(35% of the vocabulary)
descriptors per molecule	1.35	(22,929 of 25,106 carry ONE)

It is free text, not a taxonomy. sweet/sweet-like, fruity/fruit, mint/minty/peppermint/spearmint, wood/woody, sulfur/sulfury/sulfurous and jasmin/jasmine all coexist as separate labels; URL encoding leaked in (red%20rose, cut%20grass). Axes are mixed: sensory qualities (bitter), source referents (apple, bacon, taco), intensity (strong, faint) and chemical class names (aldehydic, phenolic, terpene, pyridine). The last group is circular for our purposes -- "phenolic" is a statement about structure.

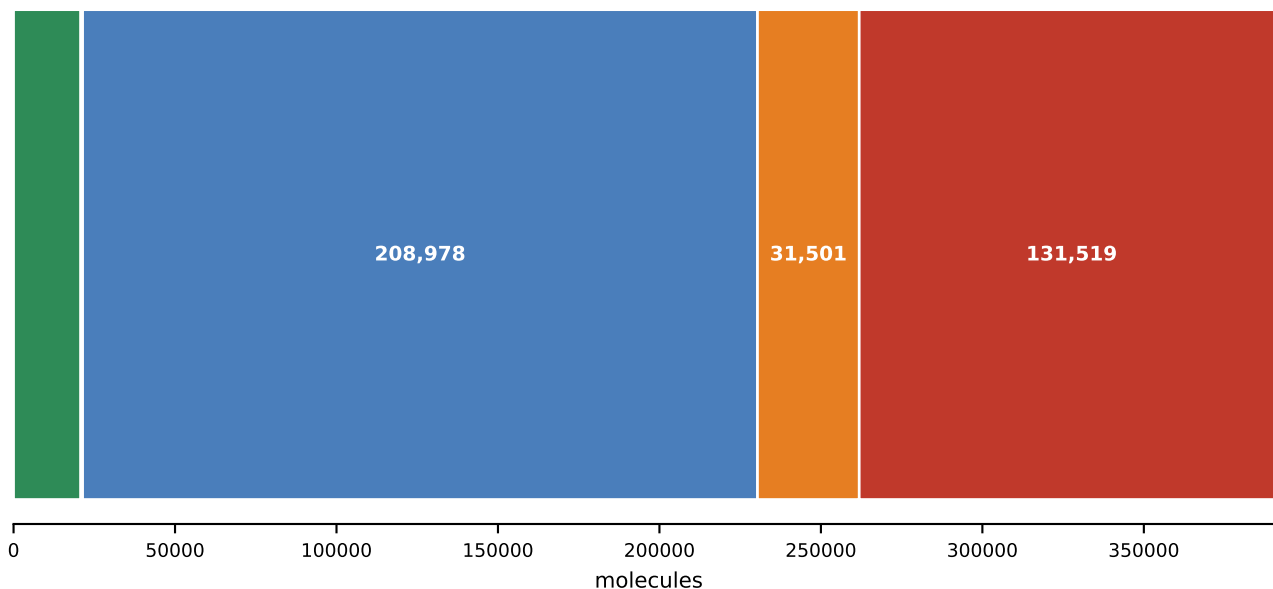
THE sweet-like PROBLEM

sweet-like is 40.8% of all assignments on its own and is almost certainly IMPUTED rather than measured. Molecules carrying it are dominated by catalogue identifiers (SCHEMBL, ZINC, CID, AC1...) -- compounds no human has tasted -- while named aroma compounds sit in the small classes. Excluding it is necessary for any honest evaluation, and doing so removes two thirds of the data.

After synonym folding into 22 sensory classes, the vocabulary covers 94.3% of genuine (non-sweet-like) assignments. That folded set is what any future labelling effort should target.

2. Coverage is the binding constraint

corpus coconut-flavordb_v6: 393,066 molecules



FlavorDB-sourced (labelled) MW>350 glycoside (needs annotation)
InChIKey match MW<=350 (needs annotation)
MW>350 non-sugar (odorless, derivable)

measured flavor label, in corpus	20,812	5.29%
COCONUT matched to FlavorDB by InChIKey	575	0.15%

total with a measured label	21,387	5.44%

A condition vector present for 5% of instances does not teach "sweet => this R-group". It teaches "has a condition => this is a FlavorDB molecule". FlavorDB and COCONUT differ sharply in chemistry, so the model would learn SOURCE DISCRIMINATION and it would look like it was working -- the same failure shape as the 97-bit fragment convec, in different clothing.

THE JOIN DOES NOT RESCUE IT

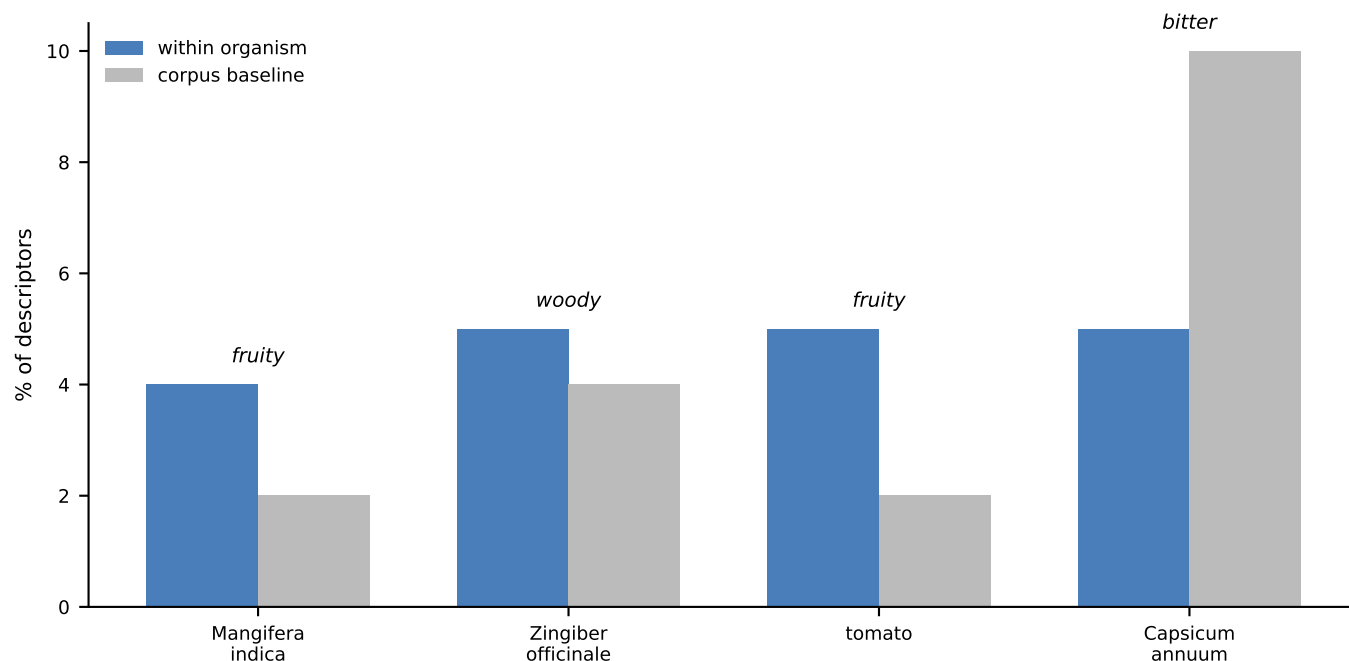
Both databases carry InChIKeys, so the join is exact and cheap:

FlavorDB with InChIKey + label	25,022	
COCONUT with InChIKey	738,827	
exact InChIKey match	5,126	0.69%
connectivity-only (skeleton)	5,729	0.78%
TOTAL reachable	10,855	1.47%

A 1.47% ceiling, and the matches are dominated by the imputed sweet-like class -- flavonoids, saccharides, triterpenoids, i.e. exactly what an automated "sugar-like => sweet" rule would catch.

3. Source organism does not predict flavor

hypothesis: herb-derived compounds taste bitter. Tested, rejected.



The reasoning is attractive: unlike chemical class or an LLM label, source organism is EXOGENOUS -- not a function of structure -- so conditioning on it could not be circular. That is a genuine point in its favour.

The data does not support it.

COCONUT compounds with organism + FlavorDB label	3,089
...with a label beyond sweet/sweet-like	1,482
distinct organisms among them	21,669

1,482 compounds spread over 21,669 organisms. And where it can be measured, the enrichment is weak or inverted: mango's compounds are 4% "fruity" against a 2% base, and Capsicum annuum is DEPLETED in bitter (5% vs 10%).

WHY IT BREAKS

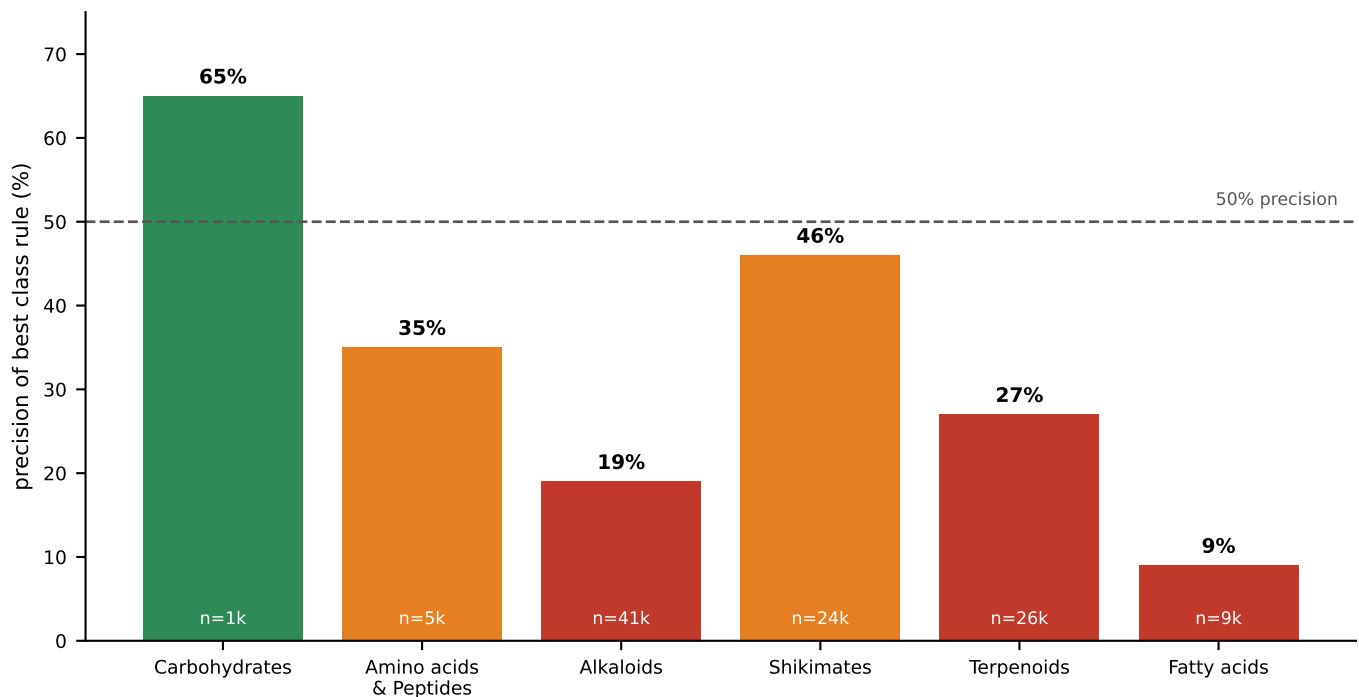
Organism -> compound is one-to-many. A mango contains ~1,500 catalogued compounds and a handful carry its flavor; the rest are sugars, sterols, structural terpenoids. "From a bitter herb" does not make a compound bitter, it makes it co-resident with one.

Two further problems: organisms cover only 37% of COCONUT, and the single largest "organism" is Homo sapiens (2,786 compounds). The field is a provenance record full of Streptomyces, E. coli and Penicillium, not a culinary one.

WHAT SURVIVES Organism is still usable as a FILTER (restrict to culinary species to raise the prior that a compound is flavor-relevant) or as a conditioning signal IN ITS OWN RIGHT ("decorate in the manner of Zingiber metabolites"), which encodes biosynthetic pathway and is leak-free. Neither requires pretending we know the taste.

4. Chemical class carries signal, but not enough to label

hypothesis: alkaloids taste bitter. True, and still unusable.



Alkaloid -> bitter is real: 19% against a 6% base rate, a 3.2x enrichment, and the chemistry is coherent throughout. Bitter peptides enrich 4.3x, fatty acids DEplete to 0.2x while showing fruity/green (they are the ester precursors of fruit aroma), carbohydrates come out 51% odorless, organosulfur compounds come out onion 9% / sulfurous 9%. None of this is noise.

It is still not a label.

targets in a class whose best rule is $\geq 50\%$ precise	1,664	(1.3%)
targets in a class whose best rule is $\geq 70\%$ precise	0	(0.0%)

"Alkaloid => bitter" is WRONG 81% OF THE TIME, and alkaloids are the single largest bucket at 41,545 compounds -- a third of the annotation load sits exactly where class tells you least.

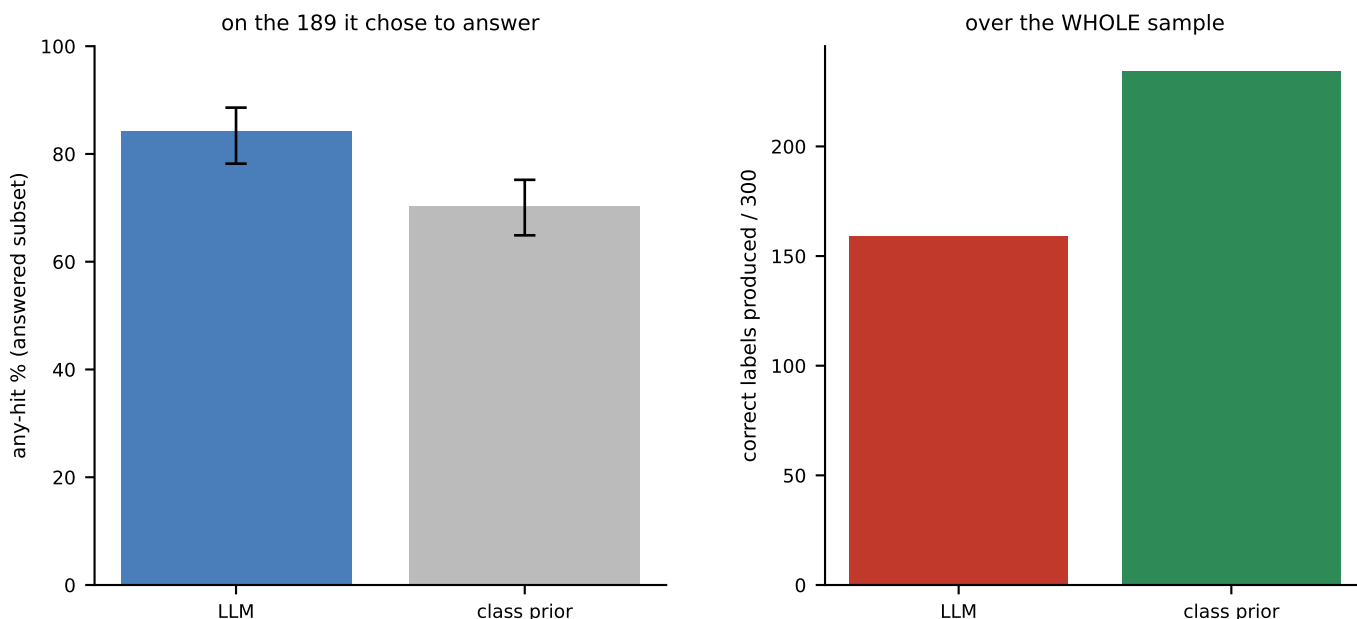
THE DEEPER OBJECTION

np_classifier_pathway is computed deterministically FROM STRUCTURE. Labelling compounds by class and then training a structure model to predict those labels teaches it "alkaloid scaffold => bitter", which NPClassifier already states exactly, for free. It is a lookup table laundered through a neural network, and the metric would look good precisely because the task is trivial. This is the same shape as the convvec leak: a structure-derived quantity conditioning a structure model.

WHAT SURVIVES Class is a good FEATURE and an excellent STRATIFIER -- report retrieval per NPClassifier pathway. It is not a flavor label.

5. LLM annotation: promising at n=20, rejected at n=300

deepseek-v4-flash, evidence-tiered prompt, 24-label constrained vocabulary



Benchmarked against 300 compounds whose flavor is measured, plus 100 unmeasured ones to test abstention. The prompt required a declared EVIDENCE TIER (documented / close_analog / structural / none) so that confidence claims are falsifiable rather than asserted.

WHAT WORKS Abstention is genuine: 83% [74.5, 89.1] on unmeasured compounds. It returned zero "bitter" on alkaloid-tagged compounds, and correctly separated amarogentin (bitter) from glycyrrhizic acid (sweet) -- both large glycosides, the circularity trap that defeats every structural heuristic. It even flagged that NPClassifier had mislabelled a synthetic Biginelli scaffold as an alkaloid.

WHAT FAILS Two things, both invisible at n=20:

the evidence tiers do not separate

documented 127/147 = 86.4% CI [80%, 91%]

close_analog 21/28 = 75.0% CI [57%, 87%]

structural 9/11 = 81.8% CI [52%, 95%]

structural is statistically indistinguishable from documented, so the tier filter -- the mechanism that was supposed to make these labels safe -- does not work. At n=20 it read 100% / 75% / 0%, from three data points.

it loses to a free baseline

head-to-head on the 189 it answered LLM 84.1% vs class prior 78.8%

over the whole 300 LLM 53.0% vs class prior 78.0%

The LLM is better on what it answers, but abstains on 37% (84 abstained, 27 no response), so the deterministic lookup produces 47% MORE correct labels for nothing. Label precision was 60.6% with recall 45.9% -- it over-predicts.

COST WAS NEVER THE ISSUE ~\$20 for 163K compounds on a cheap non-reasoning model. Reasoning models were worse AND dearer: qwen3.8-max burned 8,000 output tokens on 10 compounds and produced no answer; deepseek-v4-pro reasoned itself INTO the alkaloid class-prior trap that cheaper models avoided.

6. The glycoside trap, and the one route that works

why 'sugar-bearing => sweet' is circular, and what physics gives for free

A natural proposal: MW>350 compounds cannot be odorants, but sugar-bearing ones can still be SWEET, so label them sweet. The measured support looks decisive:

MW>350, sugar/glycoside, with a FlavorDB label	n=2,018	98.3% sweet
MW>350, no sugar	n= 186	41.4% sweet

It is an artifact. Breaking that 98.3% down by which label:

ONLY "sweet-like" (imputed)	1,863	92.3%
ONLY "sweet"	118	5.8%
not sweet at all	36	1.8%

92.3% carry only the imputed class, almost certainly assigned BECAUSE they are glycosides. The rule would read FlavorDB's own structural guess back out.

The compounds with real descriptors say the opposite:

Amarogentin	bitter	<- gentian glycoside, among the most bitter
Myronate	bitter	substances known
Neoquassin	bitter	
Swertiamarin	bitter	
Isoquercetin	odorless	
Glycyrrhizic acid	mild;spicy;sweet	<- licorice, genuinely sweet

High-MW glycosides split between intensely sweet and intensely bitter. The sugar confers solubility and receptor access, not a taste.

THE ONE ROUTE THAT SURVIVES: PHYSICAL EXCLUSION

Volatility is a hard requirement for smell -- a molecule that cannot evaporate cannot reach an olfactory receptor. That is a statement about physics, not about missing measurements.

real odorants (FlavorDB, non-sweet)	median MW 170	8.5% above 350
COCONUT	median MW 433	71.2% above 350

This yields ~209,000 honest `odorless` labels for MW>350 non-sugar compounds, at zero cost and with no model involved.

It does NOT extend to taste. Taste receptors sit in solution; steviolosides are intensely sweet at MW ~800. "Odorless" is derivable; "tasteless" is not.

AND IT MUST NOT BECOME A NEGATIVE-CLASS SHORTCUT Labelling every unlabelled COCONUT compound "tasteless" would be the classic positive-unlabeled mistake, and it is demonstrably false: 5,126 COCONUT compounds already appear in FlavorDB with real flavors. They were unlabelled in our corpus because nobody had joined the databases, not because they are tasteless.

7. Recommendation

what to do instead, in priority order

DO NOT build flavor conditioning into pretraining. There is no data foundation for it at corpus scale, and every shortcut examined either leaks (structure-derived), is too sparse (measured labels), or loses to a free baseline (LLM). The pretrained model is unconditional and that is the correct state.

THE THREE-STATE DESIGN, if flavor work resumes

positive	measured label (FlavorDB, GoodScents, BitterDB)	~21K
provable negative	physicochemically incapable of odor	~209K
unknown	small, plausible, never measured	~163K

with a `label\_source` field per molecule, and UNKNOWN MASKED OUT OF THE LOSS rather than forced to negative. Training on positives and provable negatives while abstaining on the rest is the honest version. Two states would silently convert "not measured" into "has no flavor".

RANKED NEXT STEPS

1. Use the 21,387 measured labels as an EVALUATION STRATIFIER, not a training condition. Report retrieval per flavor class. Costs nothing, needs no labels in training, and answers "is this model useful for flavor chemistry?" directly.
2. Apply the ~209K physical odorless labels. Free, honest, model-free.
3. If conditioning is wanted: pretrain unconditional (done), then FINE-TUNE conditionally on the ~21K measured subset. No imputation anywhere.
4. The one defensible LLM use is narrow: the 31,501 MW>350 glycosides, where the class prior is PROVABLY wrong (the sweet/bitter split above) and no cheap alternative exists. ~\$165 on a strong model. A separate research question, not a blocker.

WHAT TO CARRY FORWARD REGARDLESS

- * Any condition vector must pass a SHUFFLE TEST: permute conditions across instances and re-score. Unchanged retrieval means the condition is ignored; a collapse far beyond what permutation justifies means it is leaking rather than conditioning. This single diagnostic would have caught the original 97-bit condvec immediately instead of after a full seed sweep.
- * `sweet-like` must be excluded from every ground-truth set. It is imputed, it is 40.8% of the data, and including it gives credit for reproducing an imputation.
- * COCONUT's `np_classifier_*` and `chemical_class` ARE available at ~90% coverage and are reproducible and auditable. They are the honest conditioning candidate if one is wanted -- but they are `f(structure)`, so shuffle-test them before trusting any gain.